Publishing a note with an AI assistant

NSDF-Data student guide 5 — with Claude Code or a similar assistant

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Tip

This is guide 5 of 5. It shows how to use an AI coding assistant, such as Claude Code, to do the tedious parts of writing and publishing a note. It does not replace guides 3 and 4. You still own every number and every sentence.

What the assistant is good at here

- converting a Word or Google Docs draft to a note (with `tools/word_to_note.py`), and fixing the formatting it leaves behind;
- reading the check's messages and fixing the file for each one;
- checking that every notebook link is pinned to a real commit;
- finding numbers in your draft and confirming them against your notebook's output;
- checking the header, the section headings and the outline.

What it must not do

Careful

- **Never let it invent or round a number.** Every number in the note must come from your notebook output.
- **Never let it write your reading or your conclusion.** It can tidy your sentences. The conclusion must be yours.
- **Never let it edit other people's notes,** `docs/index.html`**, or anything under** `docs/`**.** Those are generated by the publishing step.
- **Never let it push to** `main` **or** `develop`**,** commit data files, or commit anything under the `idx` folder.
- **Never let it run the notebook** that overwrites `data_analysis/archives/good_noise.h5`, unless your lead has said it is fine.

Setting it up

1. Install the assistant. For Claude Code, follow Anthropic's setup instructions at docs.claude.com.
2. Open a terminal in the `NSDF-Data` folder and start the assistant there. It needs to see the repository.
3. Make sure you are on your own branch: `git checkout student-yourname`. Run `git status`, and check that it shows no changes you do not recognise.

A starting brief

Start with a brief like this, and fill in the brackets:

> I am a student writing a note for the NSDF-Data site. My draft is `[path to your .docx or .md]`, my notebook is `[path to the notebook]`, and I committed it at `[commit hash]`. Please:
>
> 1. If the draft is a `.docx`, convert it with `python tools/word_to_note.py [path to the .docx] [note-name]`, as in student guide 4, Step 1. That puts the note in `notes/src/` and its pictures in `notes/src/img/`. Keep the header block and set `id: "TBD"`.
> 2. Check every number in the text against the notebook output, and list any number that does not match or that I have not shown.
> 3. Pin every notebook link to the commit above.
> 4. Run `python tools/check_note.py notes/src/[note-name].md`. Fix each problem it reports (the structure and the pinned links), without changing my sentences except where a fix needs it, and list the numbers it says are not in the notebook output, without changing them.
> 5. Show me the diff and stop. Do not commit or push until I say so.

The checks you still do yourself

After the assistant has finished, go through these yourself:

- ☐ Open the Markdown file and read it from top to bottom. Does every sentence say what you mean?
- ☐ Pick three numbers at random and find them in your notebook output.
- ☐ Check that every picture looks the way your caption says.
- ☐ Look at the diff. Are there changes to any file outside `notes/src/`? If so, ask why, and revert them unless you understand them.
- ☐ Run `python tools/check_note.py notes/src/your-note-name.md` yourself and confirm that its last line says *ready for review*. Then look at each number it lists as not found.

Committing and opening the pull request

Once you are satisfied, follow guide 4 from Step 2 onward. You can ask the assistant to write the commit message, but read it before you commit.

When the assistant goes wrong

- **It says it has checked a number, but the number is not in the notebook.** Stop, and ask it to show you the line of output it used. If it cannot, the number is wrong.
- **It changes a sentence you wrote into a different claim.** Revert that sentence, and say so in your next message to the assistant.
- **It wants to change `docs/` or `index.html`.** Do not allow it. Those files are generated.
- **It keeps failing the check.** Ask your lead. Paste the check's message, not a summary of it.

Session Info

Guide version 5, 20 October 2026. Written for every computer. The AI tools change quickly, so if a step here no longer matches what you see, follow the assistant's own instructions and tell your lead.

Where to get help

Anthony Villano, project lead: anthony.villano@ucdenver.edu

Include the brief you gave the assistant, what it changed, and the check's last message.

Key links

- Guide 3, writing a note; guide 4, publishing a note: https://villano-lab.github.io/NSDF-Data/
- Pull requests: https://github.com/villano-lab/NSDF-Data/pulls
- Claude Code documentation: https://docs.claude.com